



Credit **Solvency** Assessment for New Financial Product Applicants

MACHINE LEARNING

DATA SCIENCE

CREDIT SCORING

FINANCIAL ANALYTICS



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Data Science | Credit Risk Analysis Project

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 [Notion Documentation](#)

Project Overview



STEP 01

Synthetic Data Generation

Python-generated `credit_solveny_dataset.csv` with realistic applicant features including income, DTI, and utilization.



STEP 02

Exploratory Data Analysis

Statistical profiling and distribution analysis across all features to uncover patterns and class imbalances.



STEP 03

Data Preprocessing & Cleaning

Handling missing values, categorical encoding, and feature scaling pipelines for model-ready inputs.



STEP 04

Model Training & Comparison

Logistic Regression, Random Forest, and XGBoost trained, tuned, and benchmarked across key performance metrics.



STEP 05

Results & Visualization

13 output figures covering ROC curves, confusion matrices, feature importance, and model performance comparisons.



STEP 06

Business Risk Insights

Actionable credit risk signals derived from model outputs to support financial decision-making and reduce default rates.

Risk Assessment: Making and Access With



AI CREDIT RISK ASSESSMENT

FINANCIAL RISK

The Business Problem



CREDIT DEFAULT RISK

New applicants lack credit history, making accurate solvency assessment difficult and increasing default exposure for financial institutions.



TRADITIONAL SCORING LIMITS

Manual review and simple rule-based scoring yield only **~0.70 AUC-ROC**, missing complex risk patterns and introducing human bias.



ML-DRIVEN ACCURACY

XGBoost achieves **~0.82 AUC-ROC**, automating credit decisions at scale while significantly reducing misclassification rates.



KEY RISK FACTORS

DTI ratio, credit utilization, and income-to-loan ratio are the strongest predictors | their interaction drives the majority of default outcomes.

"Data-driven models reduce bias and improve credit decision accuracy across all applicant segments."

Data Generation & Exploration

Synthetic dataset  Python-generated  Exploratory Analysis

SYNTHETIC DATASET FEATURES

Credit Score

Annual Income

Loan Amount

DTI Ratio

Revolving Utilization

Employment Type

Previous Defaults

Age

EDA METHODS APPLIED

Univariate Distributions

Bivariate Analysis

Correlation Matrix

Class Imbalance Detection

Density by Class



RISK FLAGGING LOGIC

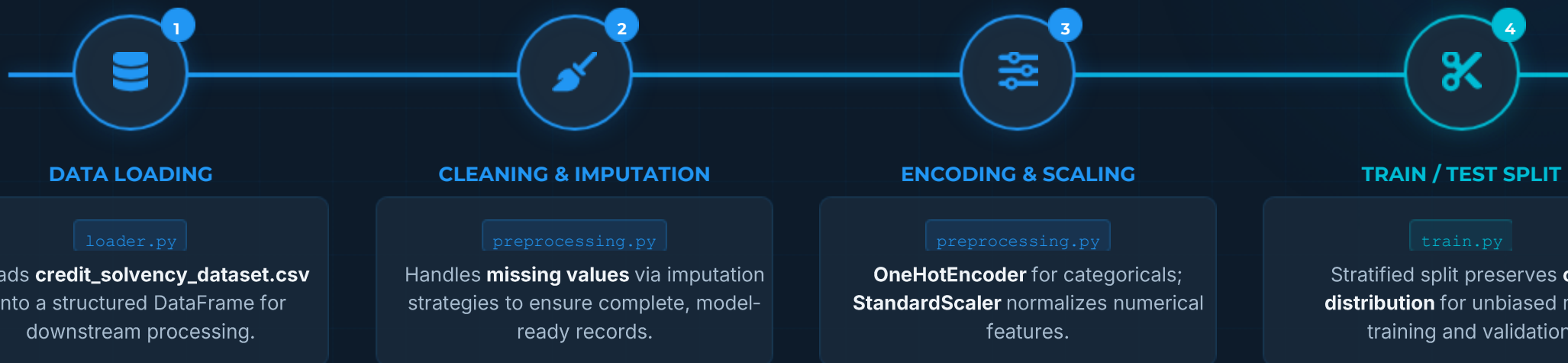
Applicants are flagged **High Risk** when **Credit Score < 600**, **> 2 Previous Defaults**, or **Income < \$40K** enabling supervised classification targets.



credit_solvency_dataset.csv

Data Preprocessing Pipeline

🔗 [src/](#) Modular Architecture



MODULAR ARCHITECTURE



Each pipeline stage is encapsulated in a dedicated [src/](#) module, enabling reuse across all four analysis notebooks.

`src/data/loader.py`

`src/data/preprocessing.py`

`src/models/train.py`

`src/visualization/plots.py`

`src/utils.py`

Machine Learning Models

CREDIT SOLVENCY PREDICTION | MODEL COMPARISON

BEST ROC-AUC

0.82

MODELS TRAINED

3

BASELINE

logistic_regression_model.pkl



Logistic Regression

Linear Interpretable Classifier

ROC-AUC 0.76

Gini Coefficient 0.52

KS Statistic 0.40

Brier Score 0.18

- ✓ Fully interpretable coefficients
- ✓ Fast training, low compute cost
- ✓ Regulatory-friendly explainability

ENSEMBLE

random_forest_model.pkl



Random Forest

Ensemble of Decision Trees

ROC-AUC 0.79

Gini Coefficient 0.58

KS Statistic 0.44

Brier Score 0.17

- ✓ Robust to outliers & noise
- ✓ Built-in feature importance
- ✓ Handles non-linear relationships

TOP PERFORMER

BEST MODEL



XGBoost

Gradient Boosting Framework

ROC-AUC 0.82

Gini Coefficient 0.64

KS Statistic 0.48

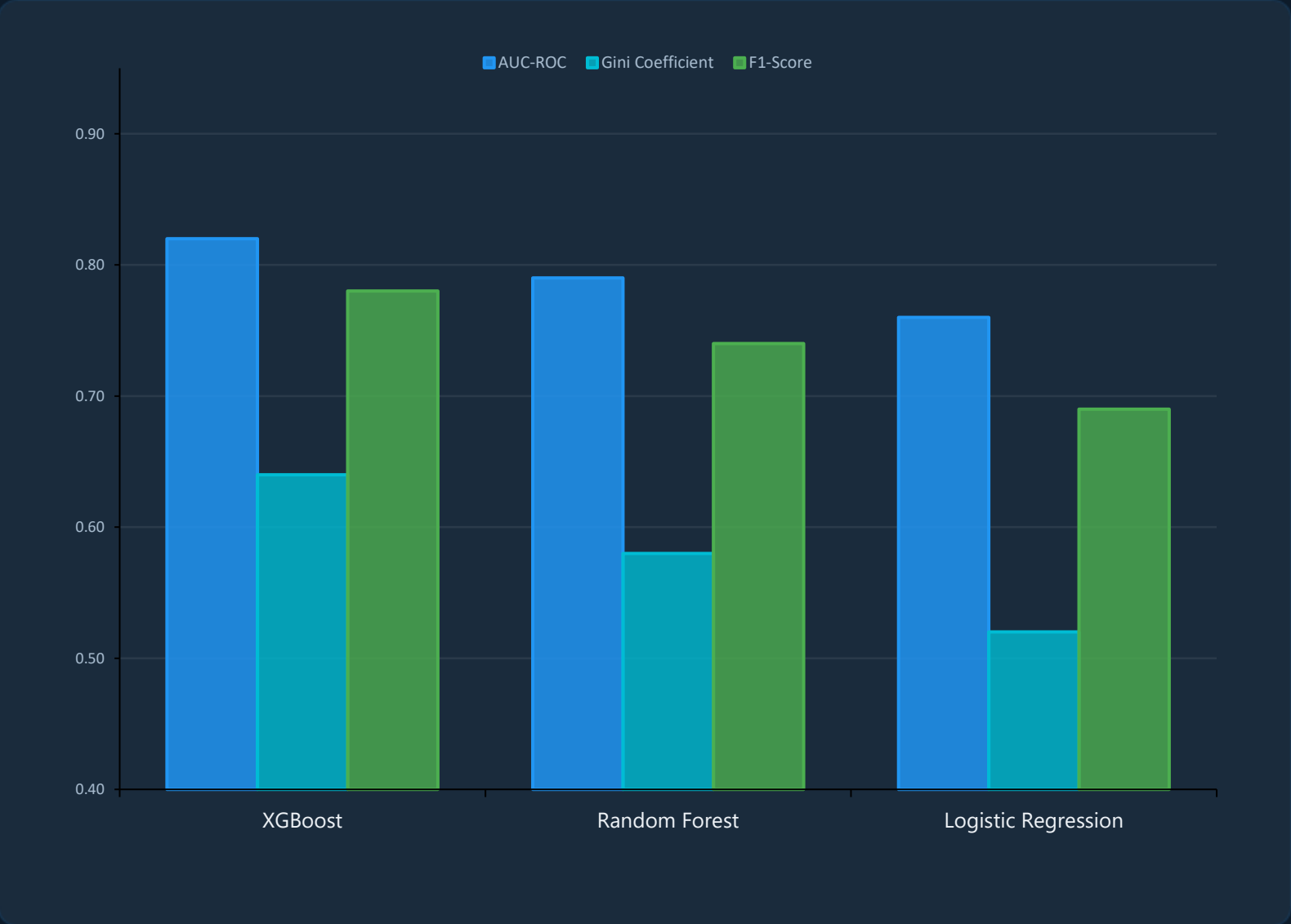
Brier Score 0.16

- ★ Highest predictive accuracy overall
- ★ Regularization prevents overfitting
- ★ SHAP-compatible for explainability

Model Performance Metrics

CREDIT SCORING EVALUATION

AUC-ROC · GINI COEFFICIENT · F1-SCORE — MODEL COMPARISON



Key Findings

ROC-AUC & GINI COEFFICIENT

XGBoost leads with AUC ~0.82 and Gini ~0.64, demonstrating superior discrimination power over RF and LR models.

PRECISION, RECALL & F1-SCORE

XGBoost achieves the **best F1 balance**, critical for minimizing false negatives in high-risk applicant detection.

CONFUSION MATRIX ANALYSIS

Confusion matrices reveal **class imbalance impact**; XGBoost correctly classifies more defaulters than RF or Logistic Regression.

THRESHOLD OPTIMIZATION

Decision threshold tuning shifts Precision-Recall tradeoff, enabling **business-aligned cutoffs** for auto-approve vs. decline rules.

KEY TAKEAWAY

For imbalanced credit data, **AUC-ROC outperforms accuracy** as the primary evaluation metric. XGBoost is the recommended production model.

OUR INVESTMENT RISK

20-49

SCORE RANGE

300 – 850

R INVESTMENT RISK TOL

CREDIT RISK FRAMEWORK

Risk Bands & Score Distribution

✓	750 – 850	Very Low Risk	AUTO-APPROVE
✓	700 – 749	Low Risk	AUTO-APPROVE
⚠	650 – 699	Medium Risk	MANUAL REVIEW
✗	580 – 649	High Risk	DECLINE
✗	300 – 579	Very High Risk	AUTO-DECLINE

KEY INSIGHT

DTI >35% + Revolving Utilization >80% is the strongest default precursor. Income-to-Loan ratio drives exposure loss 📉 these signals directly inform tier thresholds and underwriting automation.

Project Architecture & Structure

Modular | Production-Ready | Reproducible Python Codebase



data/

Dataset Storage

Houses **credit_solvency_dataset.csv** | the primary input for all model training and evaluation pipelines.



notebooks/

Analysis Pipeline

Contains **4 Jupyter analysis scripts** covering EDA, feature engineering, model comparison, and result visualization.



src/

Modular Source Code

Core modules: **data_loader**, **preprocessing**, **model_training**, **visualization**, **utils** | each with a single responsibility.



models/

Trained Model Artifacts

Stores **3 serialized .pkl files** (XGBoost, Random Forest, Logistic Regression) ready for inference and deployment.



tests/

Unit Test Suite

Includes **test_data_loader.py** and **__init__.py** to validate data ingestion and ensure pipeline reproducibility.



docs/

Project Documentation

Contains **project_overview.md** with methodology, architecture decisions, and usage instructions for collaborators.

Conclusions & Next Steps

XGBoost-powered credit risk pipeline  ready for production

KEY FINDINGS

- **XGBoost AUC 0.82** outperforms Random Forest (0.79) and Logistic Regression (0.76)
- **DTI + Revolving Utilization** interaction is the strongest default predictor
- **Risk bands** enable automated, threshold-driven underwriting decisions
- Modular Python pipeline ensures full **reproducibility** and scalability

NEXT STEPS

- Deploy model via **REST API** or **Streamlit** dashboard for real-time scoring
- Integrate **real-world credit bureau** data for improved generalization
- Add **SHAP / LIME** explainability for regulatory and fair-lending compliance
- Perform **cross-validation & backtesting** on out-of-time samples



Takeaway: The modular pipeline is production-ready  next priorities are explainability tooling and real-data validation to meet regulatory standards for automated credit decisions.

PROJECT OUTPUT

Credit Solvency Model